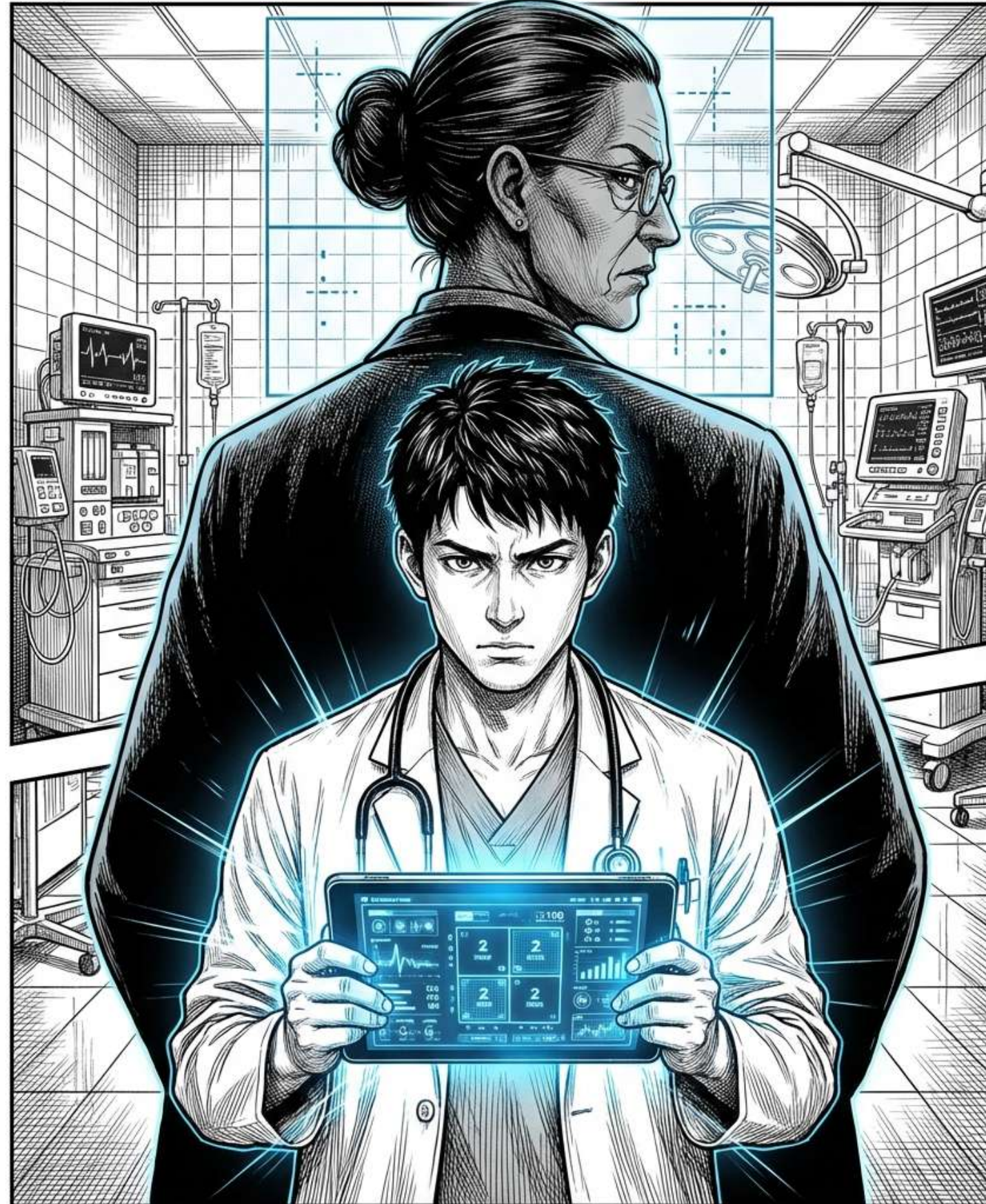
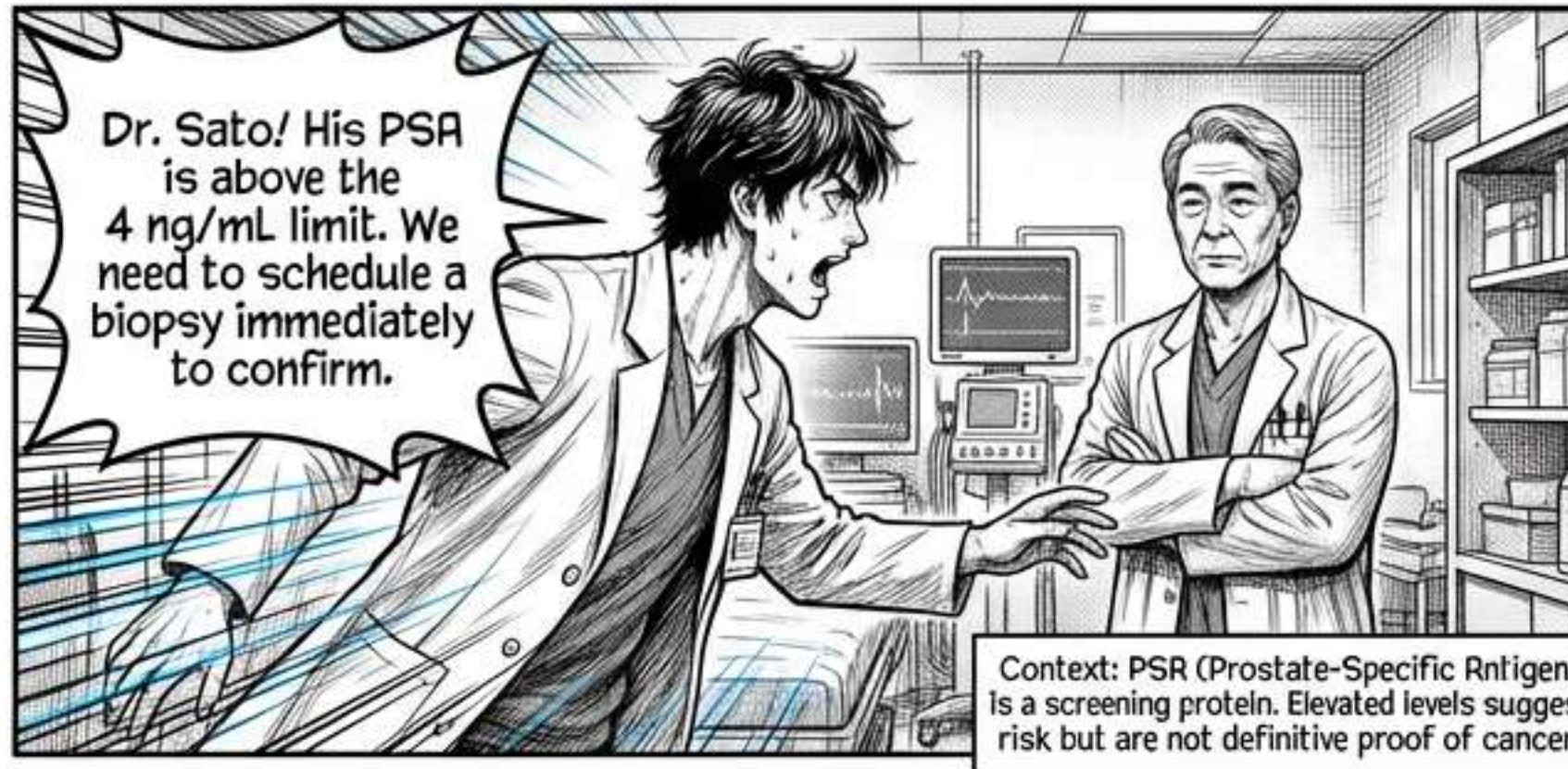


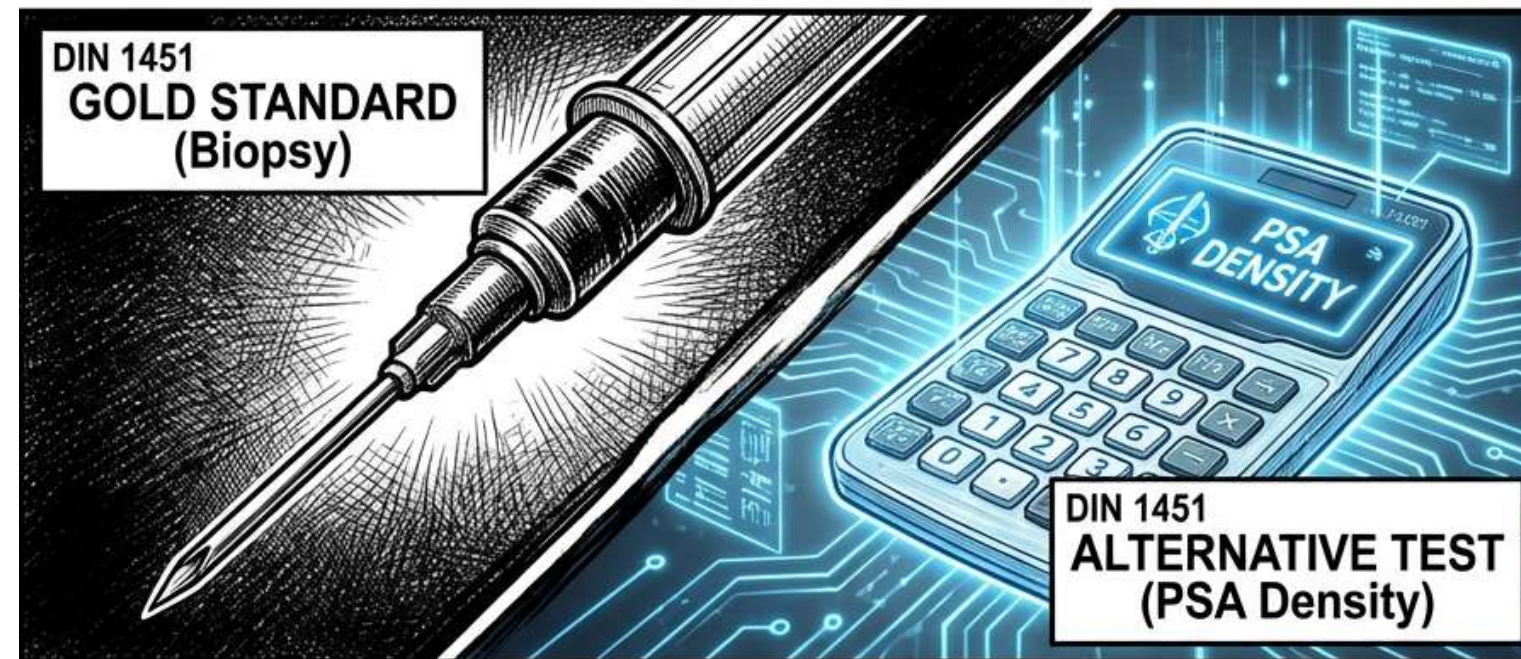
THE DIAGNOSTIC THRESHOLD

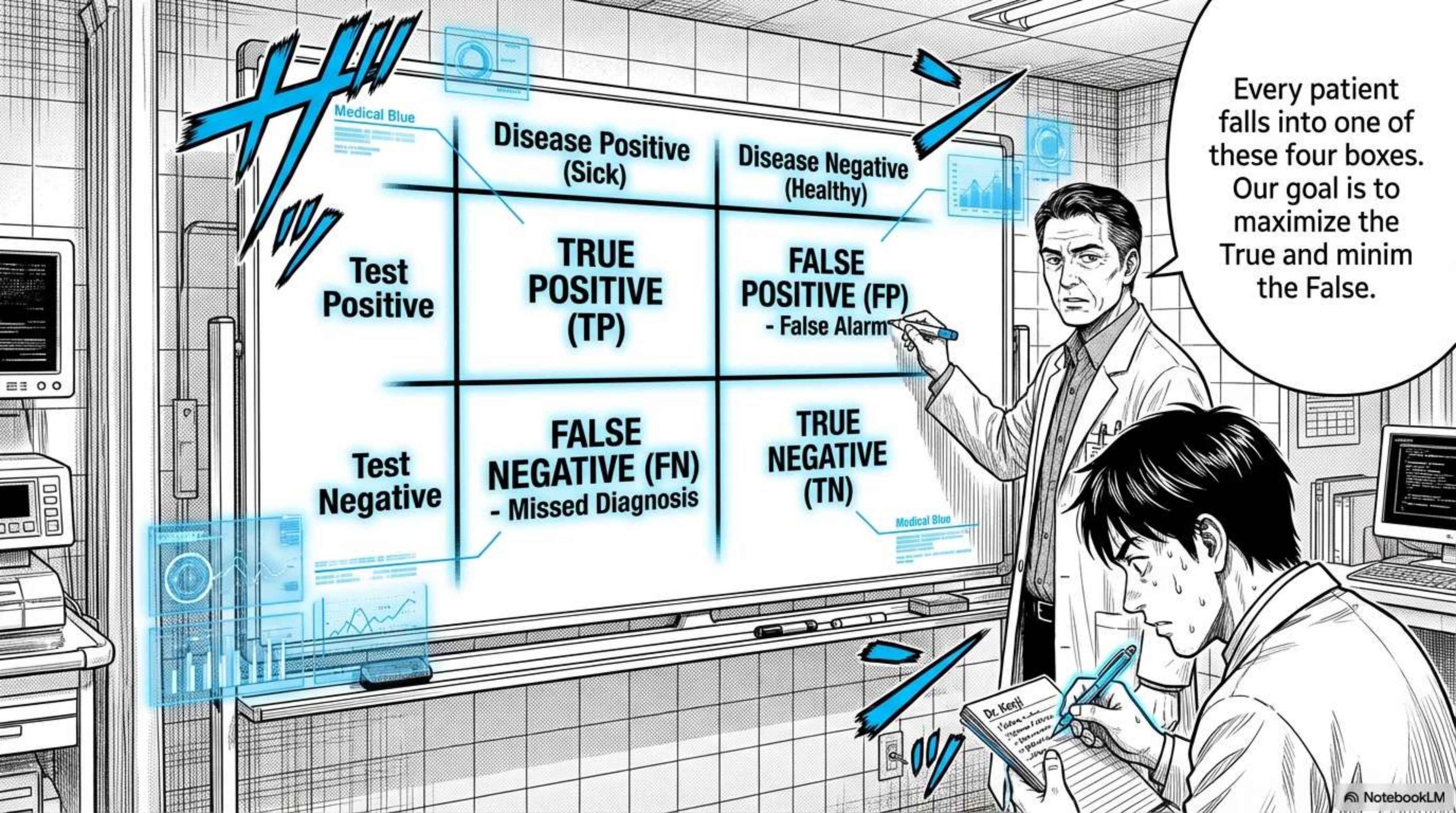
A Statistical Case Study in Sensitivity, Specificity, and Predictive Values



Based on "Foundational Statistical Principles in Medical Research" by Monaghan et al.  NotebookLM







Every patient falls into one of these four boxes. Our goal is to maximize the True and minimize the False.

Disease Positive (Sick)

Disease Negative (Healthy)

Test Positive

TRUE POSITIVE (TP)

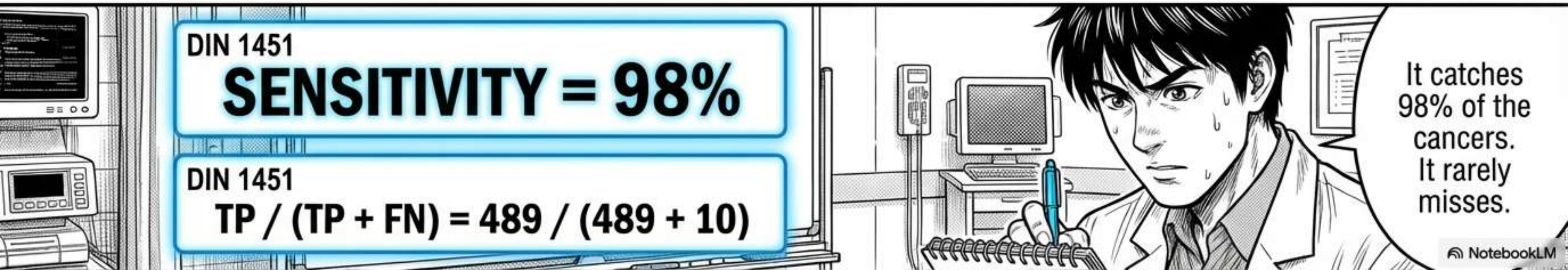
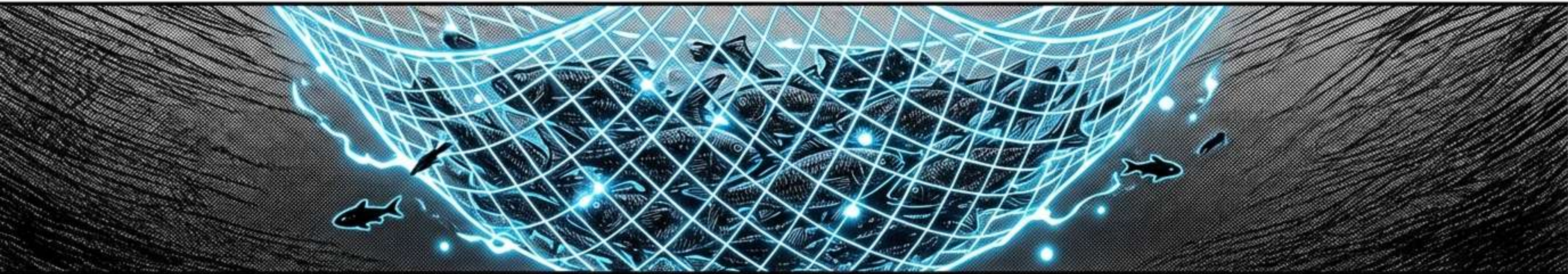
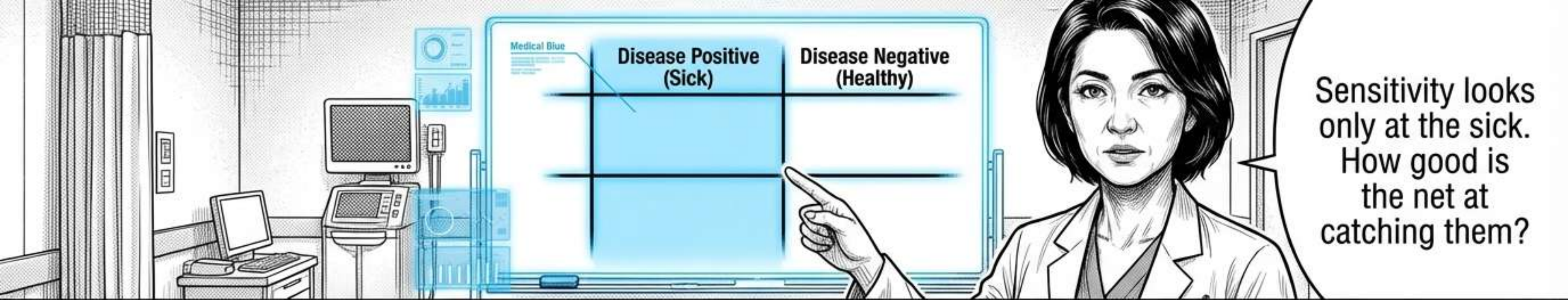
FALSE POSITIVE (FP)
- False Alarm

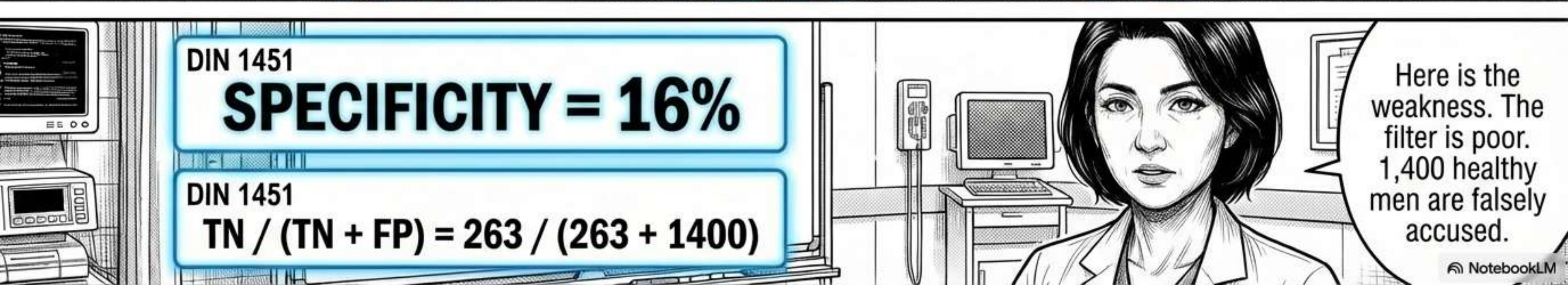
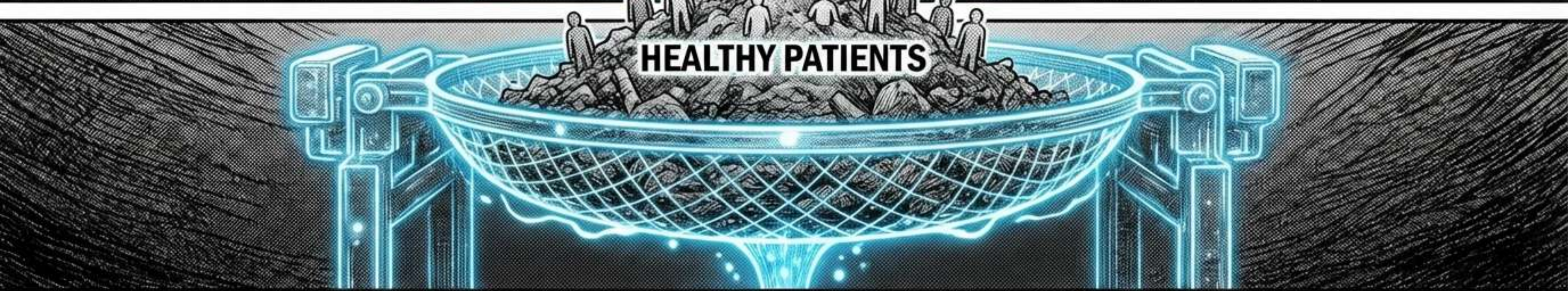
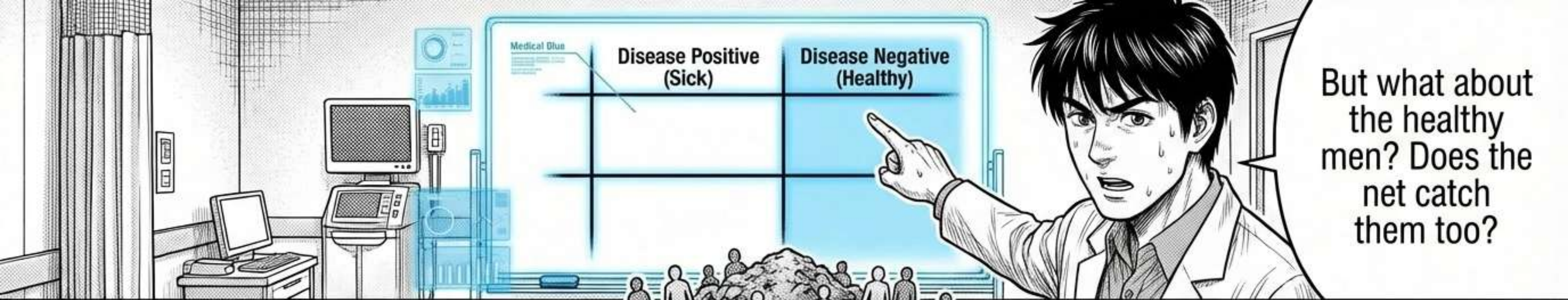
Test Negative

FALSE NEGATIVE (FN)
- Missed Diagnosis

TRUE NEGATIVE (TN)

Dr. Kenji
I have a patient
who is
very
sick





THRESHOLD (PSAD)

0.08



0.08



Cutoff 0.05

DIN 1451
Catch Everyone
False Alarm Everyone.

99.6%



SENSITIVITY

3%

SPECIFICITY

See-Saw

57%

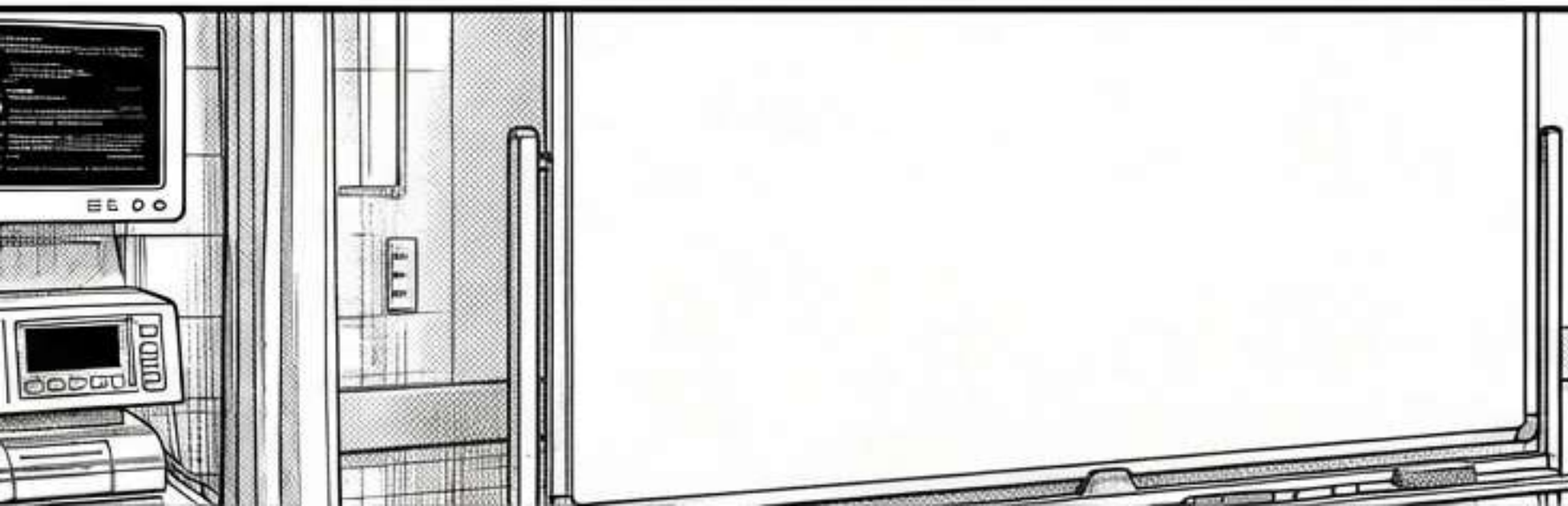
SPECIFICITY

72%

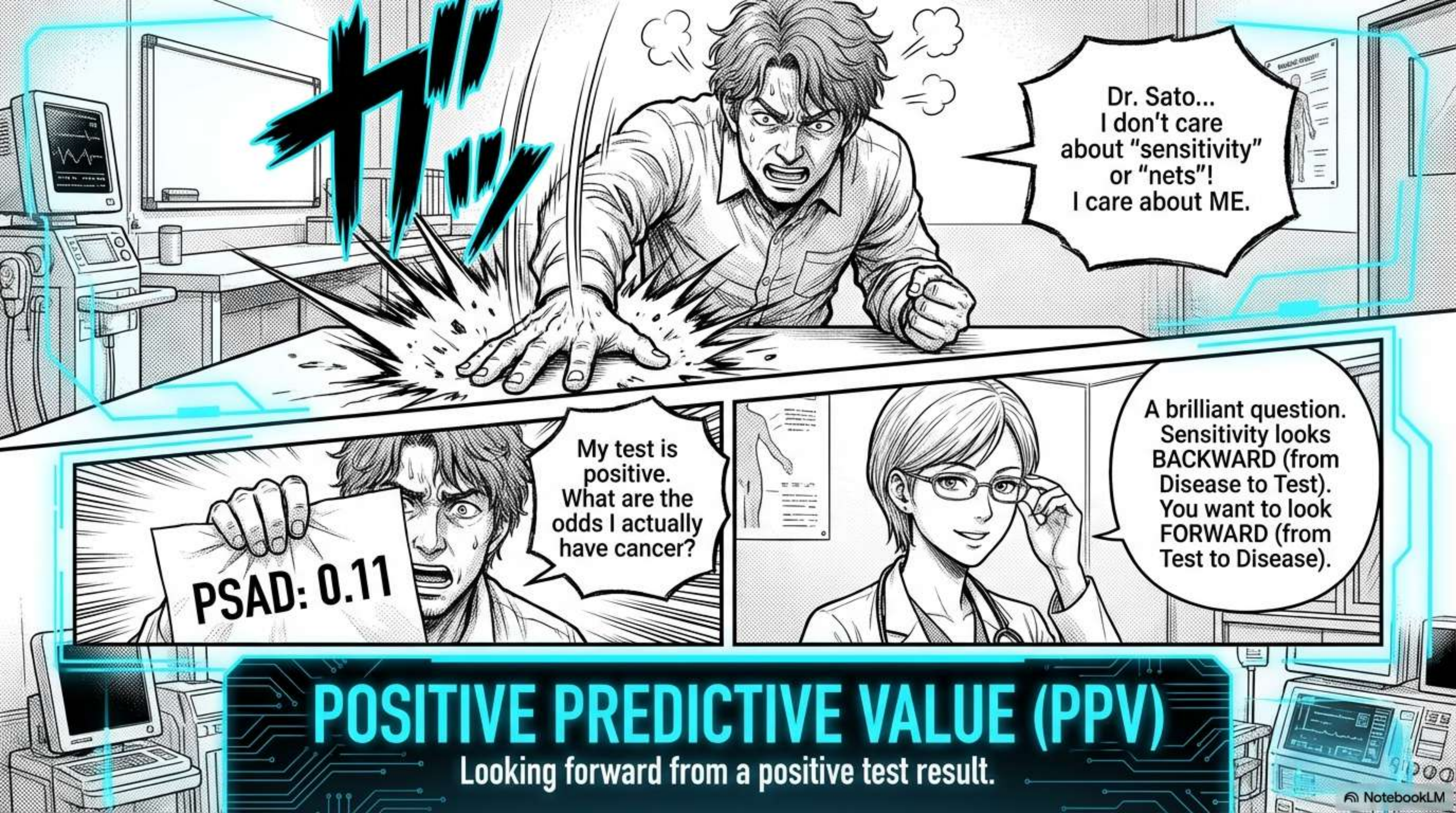
SENSITIVITY

Cutoff 0.15

DIN 1451
Better Filter
Missed Cancers.



You cannot maximize both. If you tighten the net to reduce false alarms, you will let cancer slip through.



Dr. Sato...
I don't care
about "sensitivity"
or "nets"!
I care about ME.

A brilliant question.
Sensitivity looks
BACKWARD (from
Disease to Test).
You want to look
FORWARD (from
Test to Disease).

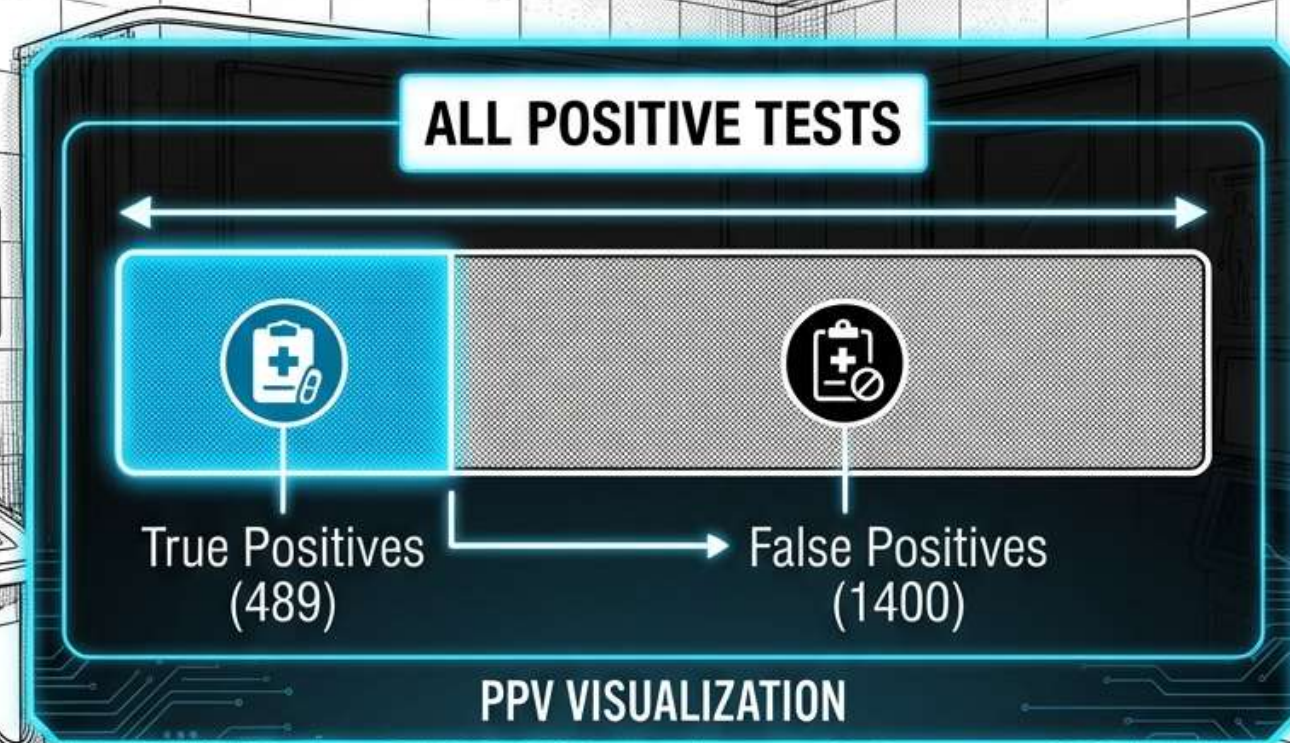
My test is
positive.
What are the
odds I actually
have cancer?

PSAD: 0.11

POSITIVE PREDICTIVE VALUE (PPV)

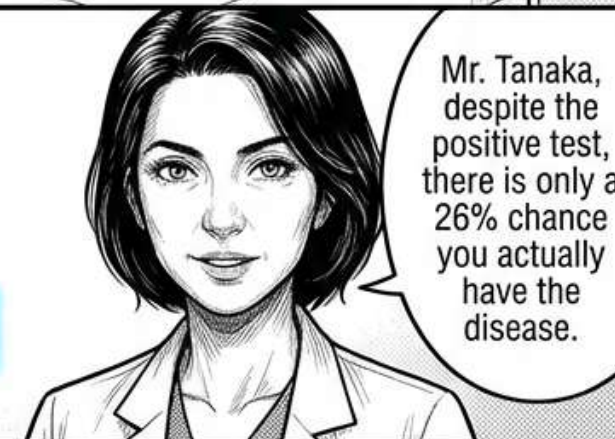
Looking forward from a positive test result.

PPV CALCULATION



$$\text{PPV} = \frac{\text{True Positives}}{\text{All Positives}}$$

$$\frac{489}{(489 + 1400)} = 26\%$$



Looking forward from a positive test result.



Test Result:
Negative (< 0.08)

ALL NEGATIVE TESTS



True Negatives
(263)



False Negatives
(10)

NEGATIVE IZATION

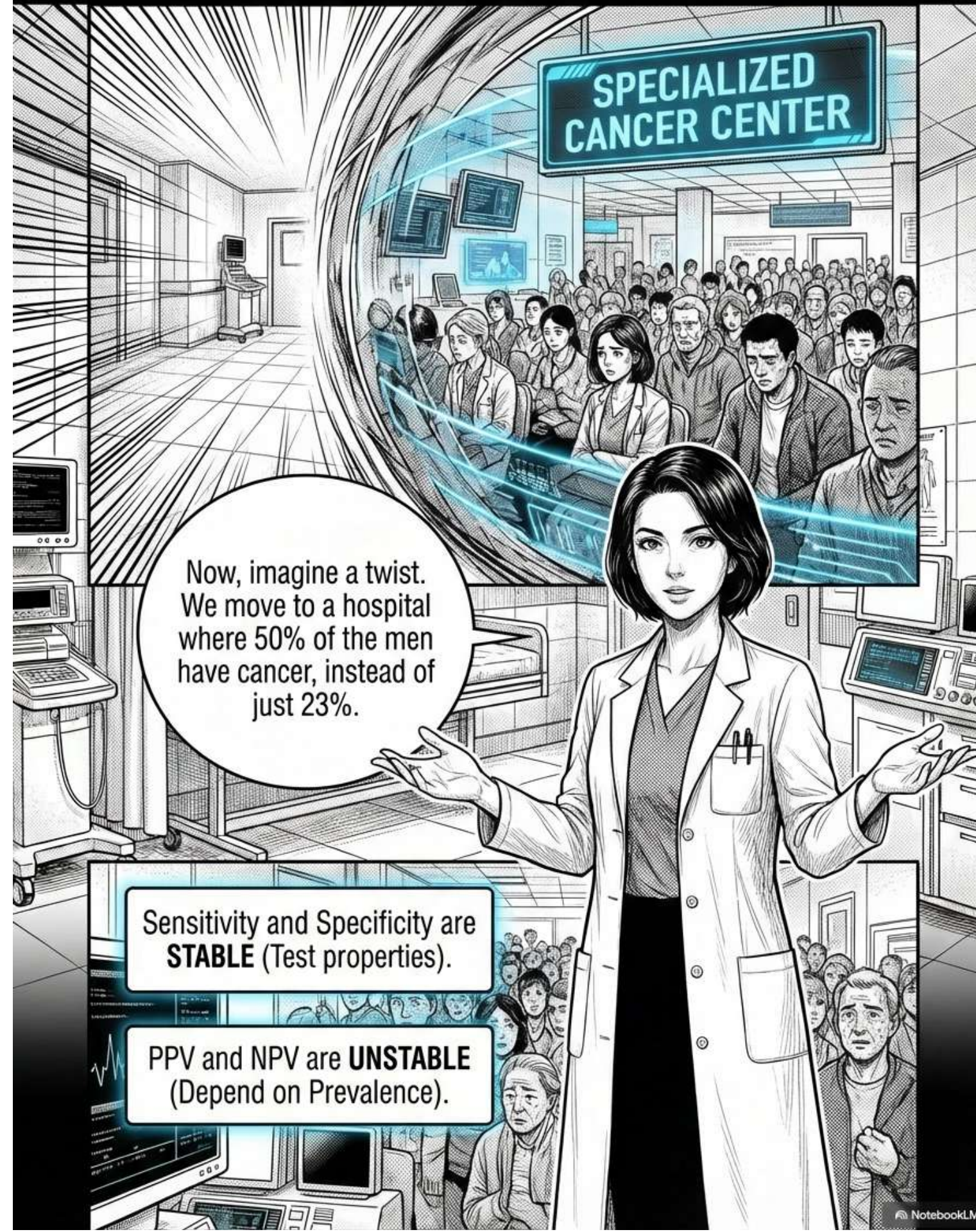
$$\text{NPV} = \frac{\text{True Negatives}}{\text{All Negatives}}$$

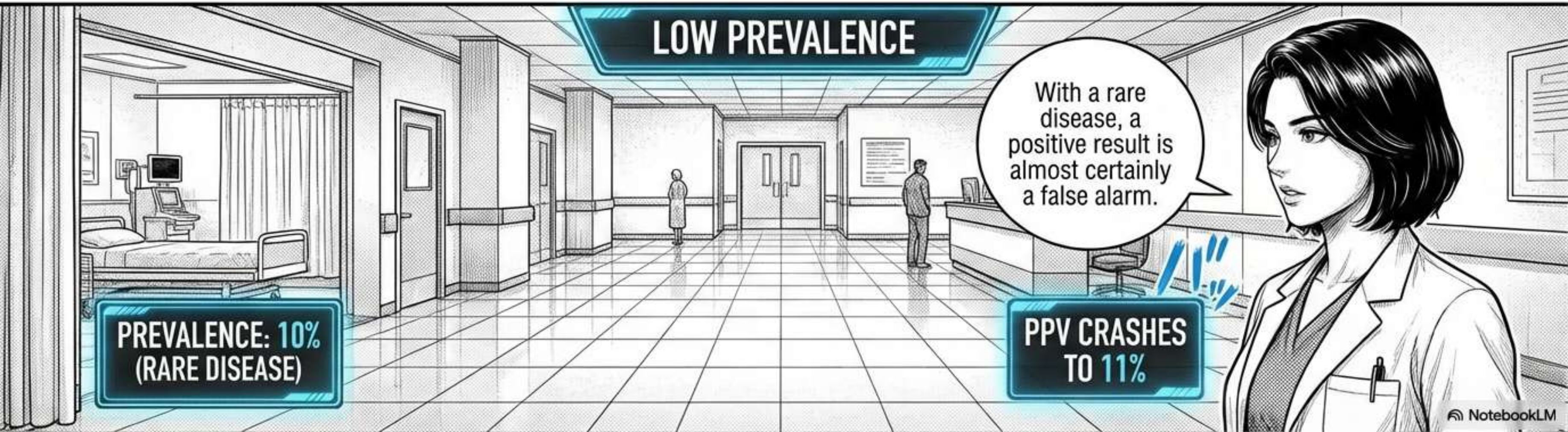
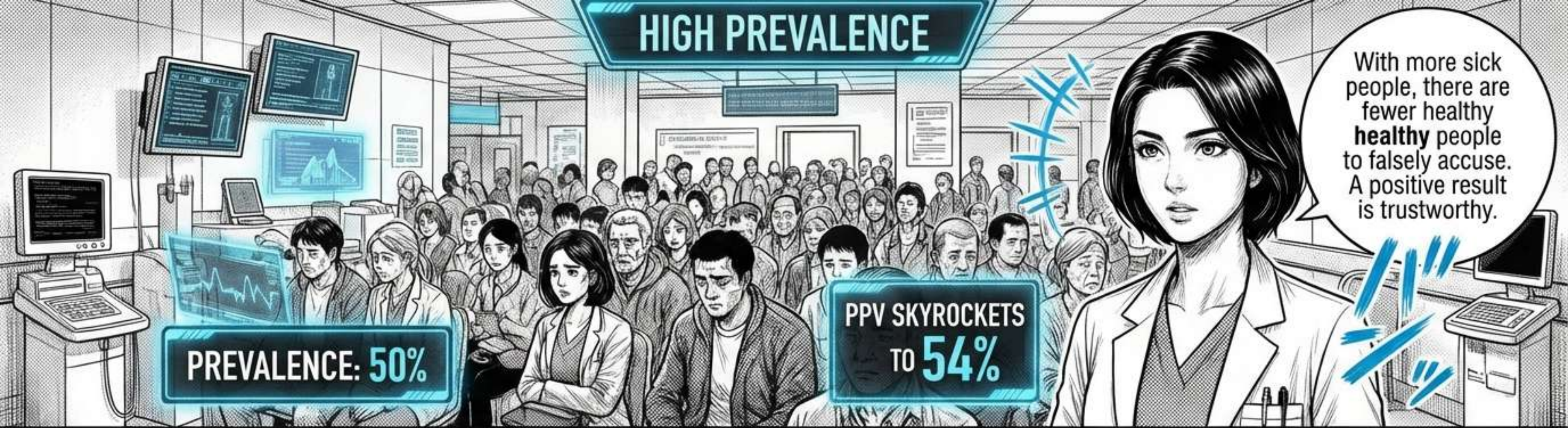
$$\frac{263}{(263 + 10)} = 96\%$$

If the test was negative, we could be 96% sure you were healthy. This test is better for RULING OUT disease.

Looking forward from a positive test result.

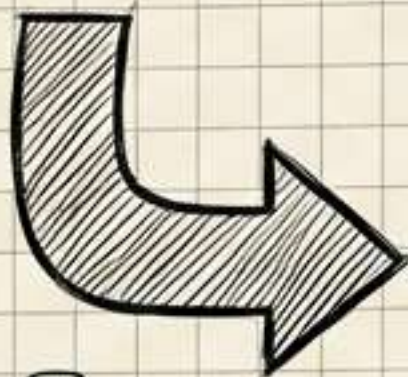
THE PREVALENCE FACTOR



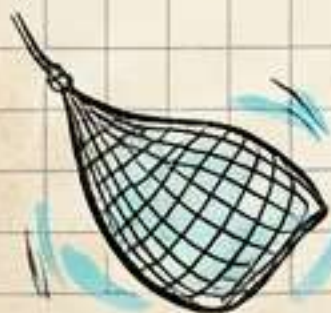


STABLE (The Machine)

- Sensitivity
- Specificity



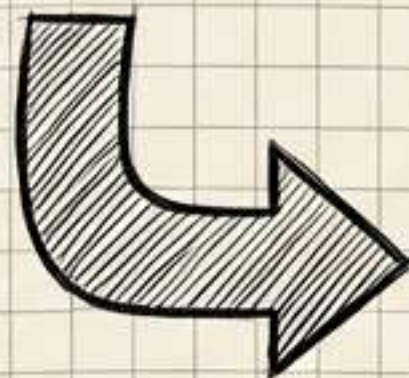
Defined by the
test cutoff.
Unaffected by
population.



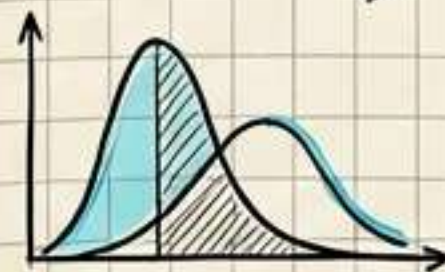
*"Sensitivity is the net.
number of*

UNSTABLE (The People)

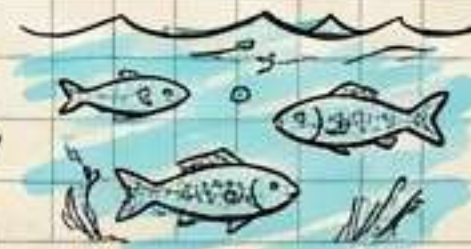
- PPV
- NPV



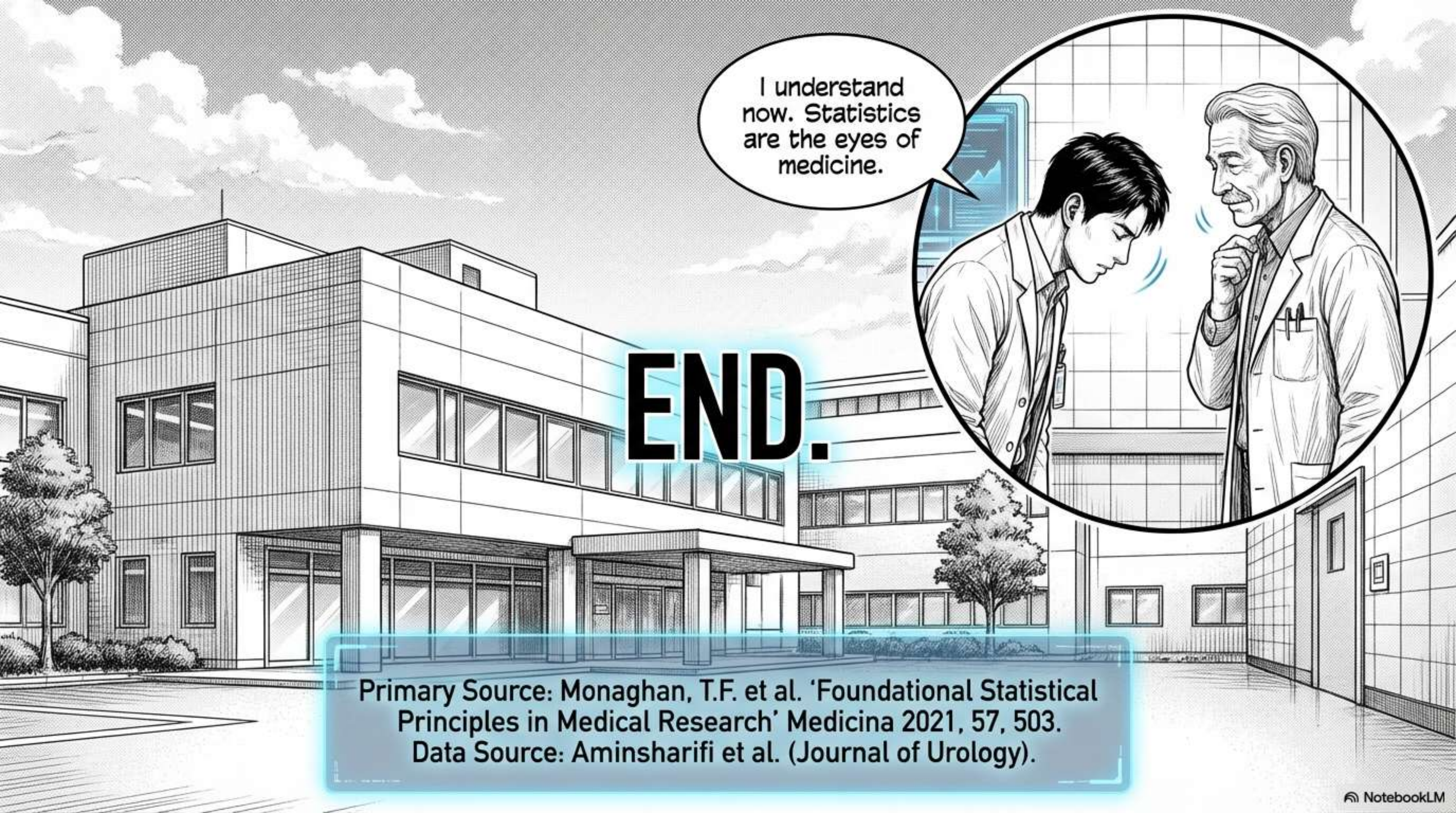
Defined by
Prevalence
(Pre-test
probability).



*Prevalence is the
fish in the water."*







I understand now. Statistics are the eyes of medicine.

END.

Primary Source: Monaghan, T.F. et al. 'Foundational Statistical Principles in Medical Research' Medicina 2021, 57, 503.
Data Source: Aminsharifi et al. (Journal of Urology).